Twelve positive unknowns for ZFC consistency

Primitive-element compression and a smaller fixed coefficient compiler

Research synthesis prepared for Vladimir Reshetnikov's ProveIt repository

22 October 2026

Abstract

We combine the repository's blank-input K215 compiler, signed Pell arguments, and root-amalgamation constructions. Four witnesses—the main-root quotient and three Pell roots—are replaced by a single integer primitive element. The result is an explicit polynomial in **twelve strictly positive unknowns**, of degree **2601**, evaluated by **167 binary arithmetic operations**. A three-equation version has maximum degree 2592 and 163 operations. The proof includes a rational-ordinate integrality lemma and explicit separation estimates for all sixteen algebraic branches; a resultant alone would not suffice. We also give unrestricted-integer variants and reduce the local compiler alphabet from 37,075 to 5,252 tiles. Compact synchronization anchors reduce the fixed base from $2^{5^{18}}$ to $2^{5^{16}}$, a factor of 25 in the bit length of $B-1$. With one coherent tuple of the four fixed coefficients, ZFC proves that absence of solutions is equivalent to Con(ZFC). This is a mathematical reduction using explicitly identified repository theorems, accompanied by exact finite checks; the new Diophantine bridge has not been formalized in Lean or Rocq. No global optimum or proof of consistency is claimed.

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1 Result, comparison, and proof boundary

The goal is an equation with *fixed integer coefficients*, whose solvability is equivalent to the existence of a finite ZFC refutation. We work at the immutable repository snapshot

`f0ae4e0991ca44a7bb4e5671d42e33d775e48035`.

The seven packages in the supplied comparison directory are numbered 09–15 [1]. They already combine a number of independent improvements, so a useful successor must specify which size criterion improves.

Theorem 1.1 (Main positive-domain result)**.** *The coefficient recipe in Section 4, and the finite arithmetic recipe in Section 2, specify a polynomial* $\mathcal{P}_{12} \in \mathbb{Z}[J, F, \alpha, z, h, i, T, s, w, \eta, \zeta, t]$ *such that*

$$\mathrm{ZFC} \vdash \left(\neg\exists (J, F, \alpha, z, h, i, T, s, w, \eta, \zeta, t) \in \mathbb{Z}_{>0}^{12}\ \mathcal{P}_{12} = 0\right) \iff \mathrm{Con}(\mathrm{ZFC}).$$

The polynomial has total degree 2601 and a division-free circuit of 167 binary operations. Multiplication by a fixed numeral is counted; reuse of an already computed value is free. The equivalent three-equation system has 163 operations and maximum degree 2592.

Here and below, "ZFC proves" refers to the ordinary mathematical proofs, internalized in ZFC and composed with the repository's calibrated K215 machine theorem. We supply no newly checked proof-assistant derivation of the Diophantine construction. In particular, the argument does not infer a ZFC theorem merely from the external soundness of Lean.

The comparison uses the reported predecessor figures, together with newly generated and checked arithmetic for the new rows. A degree for a system means its maximum residual degree.

Construction	Unknowns	Equations	Degree	Operations
Existing minimum-operation polynomial	15 positive	1	155	67
Existing weighted amalgamation (09)	13 positive	1	571	84
Existing lower-degree amalgamation (09)	13 positive	1	515	85
Existing amalgamation (09)	14 positive	1	121	76
Existing gap system (12)	15 positive	6	22	61
Existing quadratic lift (09,12)	32 positive	23	2	—
New primitive-element system	12 positive	3	2592	163
New primitive-element polynomial	**12** positive	1	2601	167
New transparent sum of squares	12 positive	1	5184	170

The reduction from thirteen to twelve unknowns is a genuine improvement within this explicit compiler family. It costs degree and operations. The 67-operation polynomial remains the preferred expression when arithmetic evaluation size is the objective; it can use the new coefficient tuple unchanged. Likewise, this work does not supersede the 32-variable quartic obtained by summing the squares of the existing 23 quadratics.

There is a separate coefficient improvement. A local pruning argument and new synchronization positions give

$$37\,075 \longrightarrow 5\,252 \quad \text{tiles}, \qquad B = 2^{5^{18}} \longrightarrow 2^{5^{16}}.$$

The exact bit length of $\beta = B - 1$ decreases from $3\,814\,697\,265\,625$ to $152\,587\,890\,625$. Both coefficient tuples remain astronomically large; the archive contains their exact finite definitions, not their expanded numerals.

1.1 The inherited interface

The four constants are $\beta = B - 1$, $K = D_C + BD_R$, M_C, and $M_F^* = M_F^{\mathrm{nat}} + B - 1$. The last port is the *shifted* mask. They must be taken from one complete, proved coefficient recipe. Their numerical mask conditions include

$$B = 2^d, \quad d \geq 4, \qquad 0 < M_C, M_F^{\mathrm{nat}} < B - 1,$$
$$M_C \equiv 2 \pmod 4, \qquad M_F^{\mathrm{nat}} \equiv 4 \pmod 8, \qquad \mathrm{pc}(M_C) + \mathrm{pc}(M_F^{\mathrm{nat}}) = d.$$

These conditions alone do not make an arbitrary tuple a compiler: the local-language, convolution, synchronization, and completeness proofs are also needed. Section 4 supplies the adapter for the new tuple.

The normalized positive core has fifteen witnesses

$$J, F, \alpha, z, f, h, i, T, s, w, \tau, \eta, \zeta, y, \gamma > 0.$$

Its first and main Pell equations, strong equation, and auxiliary equation are

$$\tau^2 - \ell(\ell + k) = 1, \qquad (X + ac + \gamma H)^2 - \Delta c^2 = 1, \tag{1}$$
$$f^2 - \Delta i^2 c^4 = 1, \qquad QV^2 - (Q - 1)y^2 = 1, \tag{2}$$

where all abbreviations are defined in the next section. The remaining conditions are the index and transport equations $N_k = N_t = 1$. The established blank-input compiler says that this core is solvable if and only if K215 halts from its blank tape.

Root reconstruction need not recover positive f, τ, y, γ. For that reason we use the signed *ordinary* core proved in package 12, `ORDINARY_BLANK_PROOF.md`. It allows $h, f, T, \tau, y, \gamma \in \mathbb{Z}$ while keeping $J, F, \alpha, z, i_{\mathrm{ord}}, s, w, \eta, \zeta$ positive. Its strong equation is

$$(i_{\mathrm{ord}} c^2)^2 = \Delta(f^2 - 1).$$

The normalized core maps into it by $i_{\mathrm{ord}} = \Delta i$. Its product theorem also permits the two outer factors to be equal signs in $\{1, -1\}$ at an intermediate stage. That theorem subsequently recovers the positive signs and the actual blank computation. This precise interface is used below; the stronger canonical restrictions "i odd" and "w a square" are needed only for the later unrestricted-integer constructions.

1.2 What the result does and does not settle

"Simplest" is not a total ordering of Diophantine descriptions. Number of unknowns, degree, circuit size, coefficient height, and proof length conflict. The new construction improves two of these criteria and supplies explicit tradeoffs for the others. Classical nine-unknown representation results already exist [10, 11]; twelve is therefore not a global record for existential representability. The contribution here is a fully specified specialization of the inspected K215/native compiler, with concrete finite coefficient data and reproducible arithmetic. Nor does the equivalence prove consistency or inconsistency of ZFC. Under the usual hypotheses, consistency prevents ZFC from proving its own consistency; a two-sided undecidability claim requires additional care about soundness and is not asserted here.

2 An explicit twelve-variable polynomial

All twelve supplied unknowns in this section are strictly positive:

$$\boxed{J, F, \alpha, z, h, i, T, s, w, \eta, \zeta, t.}$$

Every other letter denotes a substituted polynomial, not another unknown. Set

$$\begin{aligned} q &= \beta J + 1, & C &= q - F - \alpha, & X &= wq, & Y &= sq^3, \\ E &= XY, & k &= \eta + \zeta, & c &= kY + \eta, & a &= E + Y, \\ H &= 4a + 3, & \Delta &= a^2 + H = (a+2)^2 - 1. \end{aligned}$$

Define the packed rank and the two retained outer factors by

$$R = ((q-1)(q-F) + \alpha)(q^2 - 1) + (M_C + qM_F^*)J, \tag{3}$$
$$N_k = k - hE - R, \tag{4}$$
$$N_t = (K + w)C + q - F - z(q-1). \tag{5}$$

The root data are

$$\begin{aligned} \ell &= EkY, & A &= 1 + \ell(\ell + k), & I &= ic^2, \\ D &= 1 + \Delta I^2, & Q &= (\Delta I)^2, & d &= Q - 1, \\ v &= cT, & Z &= c + RD, & b &= X + ac, \\ G &= \Delta c^2 + 1, & e &= Q(Dv^2 + Z^2) - 1, & B_* &= 2QvZ. \end{aligned}$$

The letter B_* is a root coefficient; it is distinct from the fixed compiler base B.

The separating weights are also explicit polynomials:

$$\begin{aligned} g &= c + X, & u &= \ell + k, & C_* &= v^2 + D + Z, \\ r &= (a+2)I, & C_2 &= v + c + Rr, & S &= u + g, \\ \lambda &= QC_*S + 1, & \mu &= 2\lambda C_2 + S + 1. \end{aligned} \tag{6}$$

2.1 A finite norm recipe, without division or square roots

Introduce *formal computational symbols* ξ, θ in the free rank-four algebra

$$\mathcal{A} = \mathbb{Z}[J, F, \alpha, z, h, i, T, s, w, \eta, \zeta, t][\xi, \theta]/(\xi^2 - D, \theta^2 - A).$$

The fixed ports are coefficients in this display. The symbols ξ, θ are eliminated by the next finite operations and are not Diophantine variables. In this algebra set

$$\begin{aligned} \mathcal{Z} &= Ht + b - H\mu\xi - H\theta, \\ \mathcal{K} &= d(\mathcal{Z}^2 + G) - H^2\lambda^2(e - B_*\xi), \\ \mathcal{E} &= \mathcal{K}^2 - 4d^2G\mathcal{Z}^2. \end{aligned} \tag{7}$$

For $U = U_0 + U_1\theta$ write $\mathrm{N}_\theta(U) = U_0^2 - AU_1^2$, reducing $\xi^2 = D$ throughout. For $V = V_0 + V_1\xi$ write $\mathrm{N}_\xi(V) = V_0^2 - DV_1^2$. Define

$$\mathcal{R} = \mathrm{N}_\xi(\mathrm{N}_\theta(\mathcal{E})). \tag{8}$$

This is an ordinary integer polynomial. The two monic reduction rules make its computation finite and unambiguous. The delivered JSON circuits expand every addition, subtraction, and multiplication in this recipe, including all operations with fixed numerals.

The proposed single equation is

$$\boxed{\mathcal{P}_{12} := N_kN_t(1 + \mathcal{R}) - 1 = 0.} \tag{9}$$

Two useful equivalent presentations are

$$N_k = 1, \qquad N_t = 1, \qquad \mathcal{R} = 0 \tag{10}$$

and the transparent sum of squares

$$(N_k - 1)^2 + (N_t - 1)^2 + \mathcal{R}^2 = 0. \tag{11}$$

2.2 The preliminary bounds do not use the deleted roots

Suppose the outer factors are integer units. The identity

$$q - F - z(q-1) = 1 - F - (z-1)(q-1) \le 0$$

and $K + w \ge 2$ show that $C \le -1$ would force $N_t \le -2$. Thus $C \ge 0$ and $q - F = C + \alpha > 0$. The authentic mask bounds give

$$\begin{aligned} q &\le (q-1)(q-F) + \alpha \le q^2 - q, \\ 0 &< (M_C + qM_F^*)J < (q-1)(1+2q), \\ q(q^2-1) &< R < q^4 - q^3 + q^2 - 1 < q^4 - 2. \end{aligned} \tag{12}$$

Since $X \ge q \ge 16$ and $Y \ge q^3$, we have $E \ge q^4 > R + 2$. All root data that must be positive are now positive, in particular $c > 1$, $I > 1$, $D > 1$, $Q > 1$, and $d > 0$. This argument does not presuppose a main Pell root, a first Pell root, or a solution of the strong equation.

2.3 A rational auxiliary ordinate is automatically integral

Write $\chi_u(n), \psi_u(n)$ for the integer Pell sequences specified by

$$(u + \sqrt{u^2-1})^n = \chi_u(n) + \psi_u(n)\sqrt{u^2-1}.$$

Lemma 2.1 (First-coordinate divisibility)**.** *For $u > 1$ and $m, n \ge 1$, $\chi_u(n) \mid \chi_u(m)$ implies $m = (2j+1)n$ for some $j \ge 0$.*

Proof. The addition formulas give

$$\chi_u(2n + j) = 2\chi_u(n)\chi_u(n + j) - \chi_u(j),$$

and the corresponding reflection identity with j replaced by $-j$. Reduction of m modulo $2n$, followed by reflection, makes $\chi_u(m)$ congruent to $\pm\chi_u(r)$ modulo $\chi_u(n)$ for some $0 \le r \le n$. Strict increase gives $1 \le \chi_u(r) < \chi_u(n)$ if $r < n$. Divisibility forces $r = n$, which is exactly the asserted odd-multiple condition. □

Lemma 2.2 (Rational-ordinate rigidity)**.** *Let $a_0 > 1$ and V be integers. If $y \in \mathbb{Q}$ satisfies*

$$a_0^2 V^2 - (a_0^2 - 1)y^2 = 1, \tag{13}$$

then $y \in \mathbb{Z}$.

Proof. Write $a_0^2 - 1 = d_0 s_0^2$ with $d_0 > 1$ squarefree and $s_0 > 0$. For $r = s_0 y$, the number $d_0 r^2 = (a_0 V)^2 - 1$ is integral. The square of the reduced denominator of r divides squarefree d_0; hence r is integral.

Let $u + v_0\sqrt{d_0}$ be the fundamental positive solution of the integer Pell equation. Classification of its positive solutions gives

$$\begin{aligned} a_0 &= \chi_u(n), & s_0 &= v_0\psi_u(n), \\ |a_0 V| &= \chi_u(m), & |r| &= v_0\psi_u(m) \end{aligned}$$

with $m, n \ge 1$. The zero-index cases are excluded by $a_0 > 1$ and equation (13). Lemma 2.1 gives $m = (2j+1)n$. Pell composition then gives

$$\psi_u(m) = \psi_u(n)\psi_{a_0}(2j+1),$$

so $|r| = s_0\psi_{a_0}(2j+1)$ and $|y|$ is an integer. □

This special lemma is necessary: a rational root of a general nonmonic integer quadratic need not be integral. The main quotient γ will be recovered by a different integrality argument.

2.4 Separation of the sixteen branches

Over an algebraic closure of $\mathbb{Q}$, consider the tuples (γ, τ, y, f) satisfying

$$(H\gamma + b)^2 = G, \qquad \tau^2 = A, \qquad f^2 = D, \qquad dy^2 = e - B_* f. \tag{14}$$

There are exactly sixteen tuples. The last radicand cannot vanish: if $e - B_* f = 0$, then $f = e/B_* \in \mathbb{Q}$ and $f^2 = D$ makes f integral. But the definition of e, B_* would give $Q(vf - Z)^2 = 1$, impossible for integers with $Q > 1$. The same reasoning shows

$$e^2 - B_*^2 D \neq 0. \tag{15}$$

The auxiliary coordinate y may be complex during this intermediate argument. Every bound below is an absolute-value bound and remains valid for that case.

For every tuple (14),

$$|\gamma| < g, \quad |\tau| < u, \quad \sqrt{A} > g, \qquad \frac{1}{QC_*} < |y|, \qquad \frac{|y|}{\sqrt{D}} < 2C_2. \tag{16}$$

Here are the details. Because $c > 1$, $\sqrt{G} < (a+2)c$, so

$$|\gamma| < \frac{(2a+2)c + X}{4a+3} < c + X = g.$$

Also $u^2 - A = \ell k + k^2 - 1 > 0$, and $\sqrt{A} > \ell = kXY^2 > c + X$, using $c < k(Y+1)$, $X \geq 1$, $Y \geq 2$, and $k \geq 2$.

The arithmetic-geometric mean bound gives $v\sqrt{D} + Z < C_*$, hence $e + B_*\sqrt{D} < QC_*^2$. The two conjugates $e \pm B_*\sqrt{D}$ have nonzero integer product by (15), and neither has absolute value greater than $e + B_*\sqrt{D}$. Each therefore has absolute value exceeding $1/(QC_*^2)$. Dividing by $d < Q$ proves the lower bound for $|y|$. For the upper bound, $Q/d < 2$ gives

$$\frac{|y|}{\sqrt{D}} < \sqrt{2}\left(v + \frac{Z}{\sqrt{D}}\right).$$

Finally $r^2 - D = I^2 - 1 > 0$, so $Z/\sqrt{D} < c + Rr$, proving the last inequality in (16).

Lemma 2.3 (Explicit primitive-element separation)**.** *The sixteen values*

$$\gamma + \tau + \lambda y + \mu f \tag{17}$$

are pairwise distinct.

Proof. Compare distinct tuples in the order f, y, τ, γ. If f differs, its contribution has absolute value $2\mu\sqrt{D}$. The sum of all lower-coordinate differences has absolute value less than $4\lambda C_2\sqrt{D} + 2S$, which is smaller by the definition of μ. If f agrees but y differs, its weighted difference exceeds $2\lambda/(QC_*) > 2S$, whereas the remaining differences total less than $2S$. If only τ and possibly γ differ, their respective bounds are $2\sqrt{A} > 2g$ and less than $2g$. Finally the two main roots give distinct γ when every other coordinate agrees. □

2.5 Norm factorization and integral root recovery

Multiplication of the two main-root factors, then the two auxiliary-root factors, gives the following exact factorization:

$$\mathcal{R} = d^8 H^{16} \prod_{\text{sixteen tuples in (14)}} (t - \gamma - \tau - \lambda y - \mu f). \tag{18}$$

For the scale, the two γ factors multiply to $((\mathcal{Z} - H\lambda y)^2 - G)/H^2$. Multiplication over the two signs of y, with $dy^2 = e - B_* f$, and multiplication by d^2H^4 yields exactly $\mathcal{E}$ in (7). The two remaining quadratic norms give (18). In particular, $\mathcal{R}$ has degree sixteen in t, with leading coefficient d^8H^{16}.

Suppose $t \in \mathbb{Z}$ and $\mathcal{R} = 0$. Since $dH \neq 0$, one tuple has $t = \gamma + \tau + \lambda y + \mu f$. Every automorphism of the algebraic closure fixing $\mathbb{Q}$ permutes the sixteen tuples and fixes this sum. Lemma 2.3 forces it to fix the entire tuple. In characteristic zero an algebraic number fixed by all these automorphisms is rational. Thus all four recovered coordinates are rational.

Since $f^2 = D$ and $\tau^2 = A$ are integers, $f, \tau \in \mathbb{Z}$. Now $V = vf - Z$ is an integer, and the last equation of (14) becomes

$$QV^2 - (Q-1)y^2 = 1.$$

Use Lemma 2.2 with $a_0 = \Delta I$ to obtain $y \in \mathbb{Z}$. Finally,

$$\gamma = t - \tau - \lambda y - \mu f \in \mathbb{Z}.$$

This last step retains the original congruence represented by the main-root quotient; merely knowing $H\gamma + b \in \mathbb{Z}$ would not suffice.

The four deleted equations have now been restored, with possibly signed roots. Moreover $V = vf - Z = c(Tf - 1) - Rf^2$ because $f^2 = D$. Setting $i_{\mathrm{ord}} = \Delta i$ gives

$$(i_{\mathrm{ord}} c^2)^2 = Q = \Delta(f^2 - 1).$$

The signed ordinary compiler theorem described in Section 1 therefore applies. With $N_k = N_t = 1$ it proves soundness of (10). Conversely, a canonical positive normalized compiler witness gives a positive $t = \gamma + \tau + \lambda y + \mu f$ and satisfies all three equations.

2.6 The product finalizer is sound

In the rank-four algebra modulo four, $\mathcal{E} \equiv \mathcal{K}^2$. Quadratic norms are multiplicative over every commutative coefficient ring, as follows directly from

$$(ac + \delta bd)^2 - \delta(ad + bc)^2 = (a^2 - \delta b^2)(c^2 - \delta d^2).$$

Taking both norms shows that $\mathcal{R}$ is a square modulo four. Consequently $1 + \mathcal{R}$ cannot be -1.

If (9) vanishes, its three integer factors are units. Thus $\mathcal{R} = 0$ and $N_k N_t = 1$, although the two outer factors may initially both be -1. The weak bounds (12) permit both signs, so every preceding root recovery remains valid. The resulting tuple makes the full signed ordinary core product equal one. Its theorem proves blank halting and recovers $N_k = N_t = 1$ afterward. No positive sign was assumed before its justification. Positive canonical witnesses prove the converse, completing Theorem 1.1.

2.7 Uniform exact degree and arithmetic size

Assign degree one to each supplied unknown and degree zero to each fixed port. The relevant degrees are

Quantity	c	a, H	Δ	R	ℓ	D	Q	Z	e	λ	μ
Degree	5	6	12	4	11	34	46	38	122	95	116

Let a bar denote the highest homogeneous form. Then $\bar{D} = \bar{a}^2\bar{I}^2$, $\bar{Z} = \bar{R}\bar{D}$, $\bar{\lambda} = \bar{Q}\bar{R}\bar{D}\bar{\ell}$, and $\bar{\mu} = 2\bar{\lambda}\bar{R}\bar{a}\bar{I}$. In $\mathcal{K}$, the two degree-324 contributions have coefficient ratio four to one, so they leave

$$\overline{\mathcal{K}} = 3\bar{Q}\bar{H}^2\bar{\lambda}^2\bar{R}^2\bar{D}^2.$$

All terms involving a surviving formal root have lower weighted degree, with weights $\deg \xi = 17$, $\deg \theta = 11$. The subtractand $4d^2G\mathcal{Z}^2$ has degree only 392, while $\mathcal{K}^2$ has degree 648. The two norms therefore give

$$\overline{\mathcal{R}} = 3^8\bar{Q}^8\bar{H}^{16}\bar{\lambda}^{16}\bar{R}^{16}\bar{D}^{16} = 3^8\bar{Q}^{24}\bar{H}^{16}\bar{R}^{32}\bar{D}^{32}\bar{\ell}^{16}.$$

This is nonzero for every valid coefficient tuple: in particular $\bar{R} = (\beta J)^3(\beta J - F)$ is not the zero polynomial. Thus $\deg \mathcal{R} = 2592$, and $\deg \mathcal{P}_{12} = 7 + 2 + 2592 = 2601$. The unique highest square in (11) gives degree 5184. The generated circuits count respectively

$$163 = 102M + 61A, \qquad 167 = 104M + 63A, \qquad 170 = 105M + 65A,$$

where M counts multiplication and A addition or subtraction.

3 Equations over unrestricted integers

The distinction between positive and unrestricted integer unknowns is substantial here. Replacing all twelve positive unknowns by sums of squares would discard much of the improvement. Instead, we combine the new compression with the signed-domain theorem of package 14.

Lemma 3.1 (A three-variable positivity surjection)**.** *The polynomial*

$$\mathrm{Pos}(x, y, z) = 1 + x^2 + x + y^2 + z^2$$

has image exactly $\mathbb{Z}_{>0}$ *on* $\mathbb{Z}^3$*.*

Proof. The product $x(x+1)$ is nonnegative for integer x, so every value is positive. For $n > 0$, Legendre's three-square theorem represents $4n - 3$ as three integer squares. Since $4n - 3 \equiv 1 \pmod 4$, exactly one of the squares is odd. Writing that coordinate as $2x + 1$ and the others as $2y, 2z$ gives $n = 1 + x^2 + x + y^2 + z^2$. □

One copy costs seven binary operations. The signed compiler theorem allows us to retain only seven positive guards,

$$J, F, \alpha, z, s, \eta, \zeta > 0,$$

provided i is odd and w is a square. The other original witnesses may be signed. The canonical completeness proof supplies these two restrictions; they are not imposed on arbitrary old solutions.

3.1 Twenty-six unrestricted unknowns

Let h, T, t, r_0, u_0 range over $\mathbb{Z}$, put

$$i = 2r_0 + 1, \qquad w = u_0^2,$$

and replace each of the seven positive guards by its own independent copy of Pos. This uses $7 \cdot 3 + 5 = 26$ unknowns. Keep the arithmetic definitions of Section 2 through $A, D, Q, d, v, Z, b, G, e, B_*$. The old separating weights cannot simply be reused: T is signed, and $w = 0$ must still be permitted before the main root is recovered. Replace them by

$$\begin{aligned}
g &= c + X, & u &= \ell + k, & a_\tau &= 2g + 1,\\
S &= a_\tau u + g, & C_* &= v^2 + D + Z, & a_y &= QC_*S + 1,\\
r_f &= (a+2)(i^2+1)c^2, & C_2 &= c(T^2+1) + c + Rr_f, & a_f &= 2a_yC_2 + S + 1.
\end{aligned} \tag{19}$$

In the norm recipe use

$$\mathcal{Z}_{26} = Ht + b - Ha_f\xi - Ha_\tau\theta$$

and replace λ by a_y. Call the resulting polynomial $\mathcal{R}_{26}$. After the substitutions just specified, define

$$\boxed{\mathcal{P}_{26} = N_k N_t(1 + \mathcal{R}_{26}) - 1.} \tag{20}$$

Theorem 3.2. *ZFC proves that $\mathcal{P}_{26}$ has no solution in $\mathbb{Z}^{26}$ if and only if* Con(ZFC)*. Its exact degree is 5265 and its circuit has 229 binary operations. The corresponding three-equation system has maximum degree 5248 and 225 operations.*

Proof. First establish the bounds without assuming $X > 0$. The unit transport equation gives $C \geq 0$ just as before, since $K + w > 1$, and hence $R > 0$. Although X and ℓ may be zero, $Y, k, c, a, H, \Delta, D, Q, d$ are positive, with $c > 1$ and $D, Q > 1$. For each of the sixteen branches in (14),

$$|\gamma| < g, \qquad 1 \leq |\tau| < u, \qquad \sqrt{D} < r_f,$$
$$\frac{1}{QC_*} < |y| < 2C_2\sqrt{D}.$$

The bound for γ is unchanged. The inequality $u^2 - A = \ell k + k^2 - 1 > 0$ also holds when $\ell = 0$. The chosen r_f exceeds $\sqrt{1 + \Delta i^2 c^4}$ for every nonzero integer i. In the auxiliary estimates, replace v by $|v|$. We still have $|v|\sqrt{D} + Z < C_*$ and $|v| + Z/\sqrt{D} < C_2$.

The product $e^2 - B_*^2 D$ cannot vanish. When $B_* \neq 0$ the earlier argument applies. When $B_* = 0$, $e = QZ^2 - 1$ is a nonzero integer. The conjugate-product estimate gives the lower bound for $|y|$. For the upper bound, use $Q \geq 4$, $Z > 1$, and

$$|y|^2 \leq \frac{Q(|v|\sqrt{D} + Z)^2 + 1}{Q - 1} < 4(|v|\sqrt{D} + Z)^2.$$

These arguments allow signed T and i and do not involve h.

The sixteen sums

$$\gamma + a_\tau\tau + a_y y + a_f f$$

are distinct by the same successive-dominance argument. In the τ step, use $a_\tau|\tau| \geq a_\tau > g$; the stronger old inequality $\sqrt{A} > g$ is no longer necessary. The factorization has the same nonzero scale d^8H^{16}. A rational value t therefore makes all four coordinates rational. The integrality arguments of Section 2 apply, using $a_0 = |\Delta i c^2|$ in Lemma 2.2.

Only now exclude $w = 0$. The restored main equation would then give

$$(ac + H\gamma)^2 - (a^2 + H)c^2 = 1,$$

whose reduction modulo $H > 1$ is $0 \equiv 1$. Thus $w > 0$. The full signed theorem of package 14 now applies, with seven positive guards, odd i, square w, and all reconstructed coordinates integral. It gives a blank K215 halt and forces the remaining outer signs positive. The norm is a square modulo four by the same two-norm argument, so the product finalizer is valid.

Conversely take the canonical signed-domain witness supplied by package 14, with the new coherent coefficient tuple. Set $t = \gamma + a_\tau\tau + a_y y + a_f f$. This is an unrestricted integer, so it needs no sign or parity restriction. Surjectivity of Pos supplies the seven guard triples. The resulting tuple solves (20). □

The ordering of this argument matters. Invoking $X > 0$ in the separation estimates and later obtaining $X > 0$ from a recovered main root would be circular. The new a_τ eliminates precisely that problem.

After the quadratic guard substitutions, the degrees needed for the leading-term calculation are

$$\deg g = 10, \quad \deg S = 32, \quad \deg C_* = 74, \quad \deg a_y = 196,$$
$$\deg a_f = 238, \quad \deg D = 66, \quad \deg Q = 90.$$

The initial $\mathcal{K}$ has a unique degree-656 leading contribution; its competing contribution has degree 654. Successive squaring and norms give degrees $1312, 2624, 5248$. At the final norm, the competing term has degree 5184. There is therefore no highest-degree cancellation. Since $\deg N_k = 13$ and $\deg N_t = 4$, the product has degree $13+4+5248 = 5265$. The sum of squares has degree 10496.

3.2 Twenty-seven integers with a substantially lower degree

A different combination preserves the main quotient γ and eliminates only f, τ, y. Keep the same seven positive guards, let $h, T, \gamma, r_0, u_0, v_0$ be arbitrary integers, and set

$$i = 2r_0 + 1, \qquad w = u_0^2, \qquad t = 2v_0 + 1.$$

After the guard substitutions there are $7 \cdot 3 + 6 = 27$ unknowns. With the earlier root data put

$$\begin{aligned} N_m &= (X + ac + \gamma H)^2 - \Delta c^2, \\ M &= 2(D + v^2), \qquad C_a = d(t^2 + M^2 A) - e, \\ U &= C_a^2 + D B_*^2 - 4d^2 t^2 M^2 A, \\ \mathcal{S} &= U^2 - D(2C_a B_*)^2. \end{aligned} \tag{21}$$

The proposed polynomial is

$$\boxed{\mathcal{P}_{27} = N_m N_k N_t (1 + \mathcal{S}) - 1.} \tag{22}$$

It combines the weighted amalgamation of package 09 with the signed domain of package 14. Doubling M ensures the new compressed coordinate can be odd in canonical witnesses.

At a zero, N_m cannot be -1, since $\Delta \equiv 0$ or $3 \pmod 4$. Similarly $\mathcal{S}$ is a square modulo four. Thus $N_m = 1$, $\mathcal{S} = 0$, and $N_k = N_t = \varepsilon \in \{1, -1\}$. The main norm excludes $w = 0$. The weak bounds then give $E > R + 2$ and $k = R + \varepsilon + hE > 0$, so $h \geq 0$ and $R \leq k + 1 \leq 2k$. Also $Y \geq 4096$ and $c < k(Y + 1)$.

Two possible zero denominators in the proof need explicit treatment. First, $A > k^2 Y^4 > 8(c^2 + R^2)$, and

$$(D + v^2)^2 \geq 4Dv^2, \qquad Z^2 \leq 2D^2(c^2 + R^2).$$

These imply $AM^2 > 2(Dv^2 + Z^2)$. Since $d \geq Q/2$, it follows that $C_a > 0$, including for signed T and h.

Second, $B_* = 0$ would mean $T = 0$. Then $\mathcal{S} = 0$ gives $C_a^2 = 4d^2 t^2 M^2 A$. Since t is odd and nonzero, $\tau = C_a/(2dtM)$ is rational with integer square A, and is therefore integral. The integer $y = t - M\tau$ satisfies $dy^2 = QZ^2 - 1$. But $1 < Z < Q$: use $Q = \Delta(D-1)$, $\Delta > R$, and $D \geq 1 + \Delta c^4$ to get

$$Q - Z = (\Delta - R)D - \Delta - c \geq D - \Delta - c > 0.$$

Consequently

$$0 < y^2 - Z^2 = \frac{Z^2 - 1}{Q - 1} < Z,$$

whereas the left side, as a positive difference of integer squares, is at least $2Z + 1$. This contradiction excludes $B_* = 0$.

All proof-only denominators in

$$f = -\frac{U}{2C_aB_*}, \qquad \tau = \frac{C_a + B_*f}{2dtM}, \qquad y = t - M\tau$$

are now nonzero. Direct substitution yields $f^2 = D$, $\tau^2 = A$, and $dy^2 = e - B_*f = Q(vf - Z)^2 - 1$. The two rational square roots are integral, and then y is integral. Package 14 proves soundness. For completeness, its canonical c, R and auxiliary ordinate y are odd. Since M is even, $t = M\tau + y$ is odd, so the new restriction on t is available. Thus (22) has exactly the required existence predicate.

The arithmetic has 139 operations, $74M + 65A$. Its four factors have degrees $35, 13, 4, 1064$, so the exact product degree is 1116. The degree-44 terms of N_m cancel; its nonzero leading contribution is $2ac\gamma H$ of degree 35. In $\mathcal{S}$, the successive unique top contributions are dM^2A, C_a^2, and U^2, of degrees $266, 532, 1064$. This is a lower-degree alternative to the 26-integer equation, rather than a smaller-variable replacement.

Unrestricted-integer construction	Variables	Equations	Degree	Operations
Existing signed product (14)	29	1	188	121
New even-weight product	27	1	1116	139
New primitive-element system	26	3	5248	225
New primitive-element product	26	1	5265	229
New primitive-element sum of squares	26	1	10496	232

These are improvements within the specified compiler family. General eleven-integer representation results also exist [11, 12]; no global variable-count record is claimed. The attraction here is the much smaller displayed degree, finite arithmetic circuit, and concrete specialization to the supplied machine.

4 A smaller complete coefficient tuple

The arithmetic constructions above do not require a new machine. They use the supplied literal K215 table, with 215 working states, two tape symbols, and 430 transition rows. The initial state is 0, the blank symbol is 0, and the halting state is 215. The table's SHA-256 is

`fdbcef0825886035c3091c8537a69535bc96f9e265f0ae45863543d40ae70090`.

The archive includes the literal table and its authenticated transition projection. Every transition entering halt retains its original write and movement; stationary absorbing padding rules are added only after halt.

The new binding is named `pruned-T4-5252-compact-5pow16`. It modifies the four-cell local compiler in package 11 [9] in two independent ways: prune the tile alphabet, then shorten the synchronization anchor layout. We give the modified proofs here and retain the generic carry, overlap, and Pell compiler proofs as named dependencies.

4.1 Pruning while retaining every finite blank halt

Retain the 439 cell codes of T4:

- 0 and 1 are headless tape symbols;
- $2(q_0 + 1) + b_0$ is a tape symbol b_0 carrying state $0 \le q_0 \le 215$;

- $L = 434, Q_0 = 435, R_0 = 436$ are the initialized blank phases, with Q_0 carrying the initial head;
- $V_0 = 437$ and $H_0 = 438$ are vertical and horizontal boundaries.

Here Q_0, R_0, V_0, H_0 are cell names, not arithmetic registers. A tile (l, c, r, n) describes the left, central, right, and next-time central cells. Begin with the exact old T4 predicate, retained as `old_valid` in the delivered checker.

Define a horizontal edge predicate as follows. If an endpoint is V_0, the other is not V_0; a phase on its right must be L, and a phase on its left must be R_0. Otherwise $x = H_0$ if and only if $y = H_0$, and if either endpoint is a phase, the pair must be one of

$$(L, L),\ (L, Q_0),\ (Q_0, R_0),\ (R_0, R_0).$$

Apply this predicate to (l, c) and (c, r) and impose three more conditions:

1. If the triple meets V_0, it contains neither a head nor the headless tape symbol 1.
2. If $n = H_0$, the triple contains no initialized phase and no nonhalting head.
3. If $c = H_0, l = V_0$, require $n = L$; if $c = H_0, r = V_0$, require $n = R_0$.

Order the tiles with the Start tile $(434, 435, 436, 1)$ first and the distinguished forbidden tile $(438, 0, 0, 438)$ second, followed by all other retained tiles lexicographically. Keep the forbidden tile as an exception to the new filter. It satisfied the old predicate but was globally impossible: its left H_0 has an ordinary right neighbor, contradicting horizontal boundary propagation there.

Proposition 4.1 (Local-language preservation)**.** *The new marked periodic-tableau language is nonempty exactly when the literal K215 machine halts from its blank tape.*

Proof. Every retained tile is an old T4 tile. Old soundness therefore gives a blank halt; the exceptional forbidden tile cannot occur globally.

Conversely a finite blank halt visits a finite tape interval. Put the walls beyond that interval with at least three permanently unused blank cells on each side. Choose the padded width and height independently as sufficiently large powers of five. Genuine rows satisfy the edge predicate: initialized rows are $L^u Q_0 R_0^v$, horizontal boundaries propagate within their strip, and all other interior rows are ordinary tape configurations. The unused margins satisfy the new wall restriction. A row immediately preceding a horizontal boundary is a halted row, so its only head is a halt head and it contains no initialized phase. K215's first transition goes to working state 93, which also excludes an initial row immediately followed by a stop boundary. The first row after a horizontal boundary has L, R_0 next to the walls.

The helical extension shifts the adjacent strip by one time step across a wall. Permanently blank margins remain headless at both times, and V_0 blocks horizontal phase/boundary propagation. Thus all new conditions hold across seams as well. In particular, a tile centered at V_0 may see different row types on its two sides; the predicate explicitly permits this. The old construction's canonical Start and absence of the forbidden tile are preserved. □

There are 46,980 triples after the original head filter, and 5,215 after the new horizontal filters. Testing their 439 possible successors and adjoining the exceptional tile yields 5,252 tiles. Their census is

Central cell class	Count	Explanation
Headless 0 or 1	3495	1751 zero, 1744 one
Working-state head	1720	$215 \cdot 2 \cdot 4$
Halt-state head	16	$2 \cdot (4+4)$
L, Q_0, R_0	7	$3+1+3$
V_0, H_0	14	$9+5$
Total	5252	

The complete ordered list is included, with SHA-256

`baa3d2b5296b818b2e6488c1b06fda29e405d9b4cf5dd19a3c38fbdd1ba658d7`.

This bounded enumeration never asks whether K215 halts.

4.2 Exact finite formulas for all four ports

Use the fixed values

$$A_0 = 8192 = 2^{13}, \quad \rho = 2^{78125}, \quad L_{\mathrm{cell}} = 1953125 = 5^9, \quad B = \rho^{L_{\mathrm{cell}}} = 2^{5^{16}}.$$

Write the ordered tiles as (l_j, c_j, r_j, n_j), $0 \le j < 5252$. For $0 \le u \le 438$, set

$$e_l(u) = 8764 + u, \quad e_c(u) = 8325 + u, \quad e_r(u) = 7886 + u, \quad e_n(u) = 7008 + u.$$

The support $\mathcal{E}$ is the union of the selector interval $\{0, \dots, 5251\}$, these four copy intervals, the dummy positions $10520, 10521$, and the anchors

$$11839, \quad 23678, \quad 35517, \quad 82873.$$

It has 7014 elements. Define

$$m_0 = A_0 - 2 + \sum_{j=1}^{2835} A_0^j.$$

For selector positions j, let

$$v_j = 1 + A_0^{1+l_j} + A_0^{440+c_j} + A_0^{879+r_j} + A_0^{1318+n_j} + \sum_{t=1}^{4} A_0^{1756+t}.$$

For copy, anchor, and dummy positions, respectively, use

$$\begin{aligned} v_{e_l(u)} &= A_0^{1+u}, & v_{e_c(u)} &= A_0^{440+u}, \\ v_{e_r(u)} &= A_0^{879+u}, & v_{e_n(u)} &= A_0^{1318+u}, \\ (v_{11839}, v_{23678}, v_{35517}, v_{82873}) &= (A_0^{1757}, A_0^{1758}, A_0^{1759}, A_0^{1760}), \\ v_{10520} &= v_{10521} = 0. \end{aligned}$$

The required integers are the following finite sums:

$$\begin{aligned} D_C &= \rho^{1317} + \rho^{605546} + \rho^{23678} + \rho^{59195} + \sum_{e \in \mathcal{E}} v_e(\rho^{771293-e} + \rho^{937043-e}), \\ D_R &= \rho^{605107}, \\ M_F^0 &= m_0(\rho^{771293} + \rho^{937043}) + \sum_{u=0}^{438} \rho^{8325+u} \\ &\quad + \sum_{u=0}^{438} (\rho^{613871+u} + \rho^{613432+u}) + \rho^{35517} + \rho^{82873}, \\ M_C &= B - 1 - \sum_{\substack{e \in \mathcal{E} \\ e \neq 1}} \rho^e - 2\rho^{10520}, \\ M_F^{\mathrm{nat}} &= M_F^0 + 4. \end{aligned} \tag{23}$$

The four supplied constants are, together,

$$\boxed{\beta = B - 1, \qquad K = D_C + BD_R, \qquad M_C, \qquad M_F^* = M_F^{\mathrm{nat}} + B - 1.} \tag{24}$$

Every exponent is fixed. These are particular integers, not additional unknowns or variable-exponent operations. The formulas and the authenticated finite tile list specify the equation completely without printing hundreds of billions of coefficient bits.

4.3 Why the shorter anchors preserve synchronization

Let $M_0 = 11839$. The old anchor set was $\{M_0, 3M_0, 9M_0, 27M_0\}$ after appropriate layout parameters. Use instead

$$\{M_0, 2M_0, 3M_0, 7M_0\},$$

with tested targets $3M_0, 7M_0$ and bare shifts $2M_0, 5M_0$. All nonanchor support is below M_0. At the first target only the first shift hits the anchor M_0; at the second only the second shift hits $2M_0$. The bare shifts cannot reach any copy target. The low copy shift 1317 is smaller than every anchor gap and cannot create an extra anchor hit.

The target difference $4M_0$ occurs in the support difference set in exactly one ordered way, $7M_0 - 3M_0$. Differences among the lower three anchors or nonanchor support are less than $3M_0$. Differences from $7M_0$ to the other lower anchors are $5M_0$ and $6M_0$, and differences to nonanchor support exceed $6M_0$. There is no circular alternative, since $L_{\mathrm{cell}} > 2(7M_0)$. Thus an aligned radix rotation satisfying both target tests must have zero within-cell shift.

For arbitrary binary rotations, the inherited clean-band argument still applies. Here $E_{\max} = 82873$, $T_1 = 771293$, $T_2 = 937043$, and

$$T_2 - T_1 = 2E_{\max} + 4, \qquad T_2 + 2E_{\max} + 2 < L_{\mathrm{cell}}.$$

At least one clause band is clean and makes its four anchors equal one. A within-radix residue $1, \ldots, b-2$ makes all candidate radix digits even. Residue $b-1$ can produce only one odd digit, from the single possible upper-dummy spill. The two anchor targets require two odd digits, so both cases are impossible. The residue is zero, and the preceding difference argument proves whole-cell alignment.

The copy geometry is unchanged in relative positions: a shift by $3 \cdot 439$ tests next-to-central agreement, and shifts by $605107 + 439$ and 605107 test the horizontal overlaps. The high copy band is below $T_1 - E_{\max}$; the two clause bands cannot contaminate each other. The exact checker tests all 1,319 copy/anchor targets against the full support. Once alignment and overlaps hold, the same native compiler recovers the four-neighbor computation plane. The local-language proposition supplies its machine semantics.

4.4 Carries, populations, and canonical signed witnesses

The mask has $\mathrm{pc}(m_0) = 12 + 2835 = 2847$, so

$$\mathrm{pc}(M_F^{\mathrm{nat}}) = 2 \cdot 2847 + 3 \cdot 439 + 2 + 1 = 7014, \qquad \mathrm{pc}(M_C) = 5^{16} - 7014.$$

The extra one in the first identity is the low marker 4. In the second, 7013 support bits other than selector 1, and the upper-dummy bit, are removed from $B - 1$. Consequently the mask population identity and the required low congruences hold:

$$\mathrm{pc}(M_C) + \mathrm{pc}(M_F^{\mathrm{nat}}) = 5^{16}, \quad M_C \equiv 2 \pmod 4, \quad M_F^{\mathrm{nat}} \equiv 4 \pmod 8.$$

Both masks are strictly between 0 and $B - 1$.

For the inherited uniform carry bound define

$$\text{mass} = (|\mathcal{E}| + 2)\left(2\sum_{e\in\mathcal{E}} v_e + 6\right).$$

Exact manageable-integer arithmetic gives bit lengths 22893 for $\sum v_e$, 22909 for $4\text{mass} + 8$, and 36856 for m_0. The sufficient radix exponent bound is $b_{\min} = 36871 < 78125$. In particular $4\text{mass} + 8 < \rho$ and $2m_0 + 4 < \rho$. Thus the original carry-free tests remain valid for arbitrary native cells, including the permitted upper-dummy bit.

The two clause contributions cancel modulo five because their gap is 2 modulo four and $b = 78125$ is 1 modulo four. The remaining terms give

$$D_C \equiv 3 \pmod 5, \qquad D_R \equiv 3 \pmod 5, \qquad 2D_C - D_R \equiv 3 \pmod 5.$$

The inherited Boolean dummy-bit construction therefore tunes the *actual packed rank* R modulo dN, using the dummy at 10520. It does not merely tune an auxiliary formal index. The required independent power-of-five width and height padding was preserved above. The canonical origin mask test and positive raw slack are unchanged; fresh Pell witnesses are chosen after the final rank is fixed.

This also supplies the signed-domain compatibility needed in Section 3. The exponent $d = 5^{16}$ is odd, the canonical word length N is a power of five, and the canonical tuning still has $R \equiv 3 \pmod 4$. Therefore

$$q = 2^{dN}, \qquad w = 2^{R-dN}$$

has square w. Choosing the canonical auxiliary index $m = cR$ supplies an odd strong multiplier i. These are completeness choices, not claims about every solution of the arithmetic core.

Compiler	Tiles	b	L_{cell}	$d = bL_{\text{cell}}$
Existing T4	37075	5^8	5^{10}	5^{18}
Pruned, old anchor pattern	5252	5^7	5^{10}	5^{17}
Pruned, compact anchors	5252	5^7	5^9	5^{16}

The exact bit lengths of β and K become respectively 152 587 890 625 and 199 861 875 001. For old T4 they were 3 814 697 265 625 and 4 685 021 093 751. The factor 25 is exact for β; it is not asserted for every port. The new tuple can be used unchanged by the old 67-operation polynomial, so this coefficient improvement is useful independently of the new primitive-element reduction.

5 Logical calibration, reproducibility, and remaining work

5.1 The exact logical statement

Let Ref_{ZFC} mean that a finite derivation of falsity exists in the repository's calibrated effective proof calculus for ZFC. Its equivalence to the usual finite-proof definition of syntactic inconsistency is part of the inherited calibration. Define Con(ZFC) as $\neg\,\text{Ref}_{\text{ZFC}}$. The proof is the composition, formalizable in ZFC, of

$$\begin{aligned}
\exists \mathbf{x} \in \mathbb{Z}_{>0}^{12}\ \mathcal{P}_{12}(\mathbf{x}) = 0 \quad &\Longleftrightarrow \quad \text{normalized/signed Pell core,}\\
&\Longleftrightarrow \quad \text{marked native tableau,}\\
&\Longleftrightarrow \quad \text{Halt}(K215, \text{blank}),\\
&\Longleftrightarrow \quad \text{Ref}_{\text{ZFC}}\,.
\end{aligned}$$

Negating this equivalence gives the requested consistency statement. The integer variants have the same last three links.

The literal Lean endpoint included for identification states

```
BlankHalts machine215 ↔ BProv ZFCax_s [] Form.fBot.
```

The pinned verification directory reports completed Lean and Rocq checks of this exact machine, with their own source, dependency, and toolchain records [2]. We did not rerun those builds. The ordinary calibration note explicitly proves the corresponding equivalence by reasoning formalizable in ZFC. We use that mathematical calibration, not an inference from the external soundness of a proof assistant to an internal ZFC theorem.

The new Diophantine part is an ordinary mathematical proof. The archive's numerical and algebraic checks are finite diagnostics and artifact checks; they are not kernel proofs of the new theorem. No accepting K215 run, ZFC refutation, or proof of Con(ZFC) is produced. In particular, the equivalence should not be described without qualification as a proof that both directions of the equation's solvability question are undecidable in ZFC.

5.2 Explicit proof dependencies

The following list separates the new arguments from the inherited mathematics that they use.

1. **Machine calibration.** The exact-table K215 endpoint and the ordinary calibration identify blank halting with a finite ZFC refutation [2, 3].

2. **Native coefficient compiler.** The fixed raw universal 78-operation proof supplies the carry-free local tests, clean bands, and rotation framework. Its 76-operation successor supplies actual-rank Boolean dummy control. The new proof replaces the alphabet and the anchor-specific synchronization argument, then checks all finite numerical hypotheses [4].

3. **Blank input and Pell bridge.** The blank-helical and 67-operation proofs remove the input, preserve a unique Start, and connect the packed word to the normalized arithmetic. The half-binomial kernel and signed Pell results remain substantive mathematical dependencies [5, 6].

4. **Signed domains.** The twelve-positive-unknown proof uses package 12's ordinary signed theorem with $i_{\mathrm{ord}} = \Delta i$. The unrestricted-integer proofs use package 14's narrower, seven-positive-guard theorem with odd i and square w [7].

5. **New compression.** The sixteen-branch separation, rational-ordinate integrality, recovery of the main quotient, exact norm, and leading-term analysis are proved in this article. The 27-integer construction adapts package 09's three-root amalgamation with an even weight and a new signed denominator analysis [8].

An especially delicate inherited point is the ordinary strong equation

$$(i_{\mathrm{ord}} c^2)^2 = \Delta(f^2 - 1).$$

One cannot simply cancel Δ and assert the normalized Pell divisibility. The relaxed Pell lemma uses a large-rank bound, strong divisibility in the integer-coefficient quadratic ring, and a proper-divisor estimate before recovering the required index relation. The inherited arithmetic proof establishes its large-rank hypothesis. Both that proof and the relaxed lemma are bundled for inspection.

5.3 What the executable checks establish

All new programs use Python's standard library. Their diagnostics cover four distinct tasks:

- **Literal data and finite compiler geometry.** Authenticate all 430 transitions, enumerate the pruned tile list, verify its census and hash, exhaust the copy/anchor target conditions, and verify exact carry, population, and five-adic inequalities. Eighteen halting control tableaux exercise 1,143 local windows. These use control transitions, not an alleged halting run of K215.
- **Circuit integrity.** Check every gate dependency, live input and operation count. The delivered JSON uses only binary addition, subtraction, and multiplication; the final comparison is to zero, or the specified constants for a system.
- **Independent algebra.** Check the norm against the product of all sixteen branches over finite fields, including encoded-root fixtures, and verify the finalizer identities and modulo-four property. The 26-integer checks include the signed and zero-scale cases relevant to its altered proof. The 27-integer circuit is also compared with a separately written direct formula.
- **Degree corroboration.** Exact univariate specializations over finite fields recover the claimed degrees. They complement the uniform leading-form proofs; alone they would not prove noncancellation for every valid coefficient tuple.

Independent mathematical reviews are included for the 12-, 26-, and 27-variable constructions. These are ordinary reviews by separate automated research workers, not independent theorem-prover certificates.

5.4 Artifact layout and reproduction

The package's entry point is `README.md`; `INSTANCE.json` binds the arithmetic to one whole coefficient recipe. `SOURCE_PINS.json` records the immutable repository paths and SHA-256 values of the bundled inherited files. `SHA256SUMS` records the delivered package contents.

The principal artifacts are:

Path	Contents
`report.tex`, `sections/`	Complete source of this article.
`report.pdf`	Compiled, visually inspected article.
`arithmetic/primitive12/`	Generator, independent checker, and the three twelve-positive-variable circuits.
`arithmetic/integer26/`	Signed primitive-element generator, independent checker, and three unrestricted-integer circuits.
`arithmetic/integer27/`	Even-weight generator, parent circuit, 27-integer polynomial, and verification receipt.
`arithmetic/baseline67.json`	The inherited minimum-operation polynomial, usable with the new tuple.
`compiler/`	Complete coefficient recipe, tile list, literal machine data, and finite checker.
`proofs/`, `inherited/`	New proof notes and reviews, plus source-pinned mathematical dependencies.
`machine/`, `licenses/`	Literal endpoint identification, machine table, and inherited attribution.
`verification/`	Reproduction driver and consolidated checks.

From the package root, run

```
python3 verification/check_package.py
make verify
make pdf
```

The first command checks the delivered data and circuit structure. The second regenerates circuits and reruns the finite semantic/algebraic diagnostics; a separate `make degrees` target reruns the slower degree corroboration. PDF compilation requires a standard LaTeX installation with the packages listed in `report.tex`. No Lean or Rocq build is included in these targets. The huge four coefficients are never expanded. The auxiliary residue evaluator computes their values modulo any requested positive integer from the same exact finite recipe.

The formulas remain valid with one of the original complete coefficient tuples when its compiler hypotheses apply. That fallback can isolate the new algebraic argument from the new coefficient adapter during review. Partial replacement of ports has no such interpretation.

5.5 Promising next reductions

The present package supports several concrete next steps. First, sharpen the separating weights. The factor of sixteen in the primitive norm is structural for these four binary root choices, but the very high *total* degree mostly comes from conservative weight growth. A smaller dominating weight could improve degree and operation count without changing the recovered variables.

Second, examine whether one of the seven positive guards in the signed theorem can be removed. Each safely removed positivity guard saves two unrestricted integers after the three-variable substitution. This needs a new sign-domain proof, not just a circuit change.

Third, optimize the local alphabet and anchor positions together. The compact pattern proves that the old geometrically growing anchor set was unnecessary. More aggressive pruning must retain a completeness padding argument and preserve the actual-rank tuning congruences.

Fourth, formalize the new bridge in manageable layers: the rational Pell-ordinate lemma; finite rank-four algebra and its factorization; separation and integral recovery; then the coefficient adapter and its connection to the existing K215 endpoint. The exact domain distinctions and the zero-scale order in the 26-integer proof are natural first formalization targets.

For incorporation, the defensible claims are therefore a reduction from 13 to 12 positive unknowns in this explicit family, from 29 to 26 unrestricted integers with a useful 27-integer tradeoff, and a 25-fold reduction in the bit length of the base-minus-one coefficient. The old 67-operation expression and existing low-degree systems remain valuable members of the comparison.

References

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[2] ProveIt contributors, *Formal correctness of the exact 215-state machine*; Lean theorem `KEndpoint215.blank_halts_iff_zfc_refutation`. Pinned verification records. Bundled endpoint and status under `machine/` and `inherited/`.

[3] ProveIt contributors, *K215 ordinary consistency calibration*, `candidate-k215/proof.md`. Pinned calibration proof. Bundled as `inherited/K215_CALIBRATION_PROOF.md`.

[4] ProveIt contributors, *Fixed raw universal 78- and 76-operation compiler proofs*; *Five-adic dummy control*. Pinned 1980-paper development directory. The corresponding full proof notes are bundled under `inherited/`.

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[6] ProveIt contributors, *Pell kernel half-binomial 42*; *Relaxed auxiliary Pell proof*; *Signed Pell proof*. Pinned native-stream-queue development. Exact source paths and hashes, including the related 1980 notes, are recorded in `SOURCE_PINS.json`.

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[12] Jonas Bayer, Marco David, Malte Haßler, Yuri Matiyasevich, and Dierk Schleicher, *Diophantine equations over $\mathbb{Z}$: universal bounds and parallel formalization*, arXiv:2506.20909v2, 2025. https://arxiv.org/abs/2506.20909v2.